

Lab 18: Evolution — Tragedy of the Commons and Game Theory (Fishing Simulation)

BIOL-8

Name: _____ Date: _____

Group:

Purpose

You will work in a **group** to fish a **shared stock** of four kinds of fish over **three seasons**.

The activity connects **evolution and ecology** to **human behavior** in two ways:

- **Tragedy of the commons** — a shared resource can be depleted when many people pursue short-term private gain.
- **Game theory** — your payoff depends on others' choices; **repeated** seasons change what looks like a good strategy.

After each season, fish that were **not** caught can **reproduce** using a fixed rule. The pool can **recover** or **collapse**, depending on what the group harvests.

Learning objectives

By the end of this lab, you will be able to:

- **Explain** the tragedy of the commons in your own words and relate it to a **shared** fish pool your group managed.
- **Describe** how **rotating** who fishes each season and **rising prices** can change incentives from one round to the next.
- **Apply** the reproduction rule to update population sizes **per species** after a season.
- **Connect** the simulation to **game theory** terms where appropriate: strategy, payoff, cooperation vs. short-term gain, and effects of **repeated** interaction across seasons.
- **Compare** the **four species** by value: gold vs. silver and large vs. small.

Background

Tragedy of the commons

When a **resource** (pasture, fishery, clean air) is **open to many users** and no one person pays the full **future** cost of overuse, each user can be tempted to take as much as possible **now**. If everyone does that, the **shared stock** shrinks and may **collapse**—even though **no one** wanted that outcome. **Rules, quotas, catch limits, and community agreements** exist partly to avoid that outcome. This lab is a **toy** version of that tension.

Game theory in one paragraph

Game theory models choices when your outcome depends on **others'** choices. In a **repeated** setting (here, **three seasons**), what pays off in season 1 can differ from what pays off after others respond in seasons 2 and 3. You may see **defection** (take a lot) vs. **cooperation** (leave fish for the group and the future) or a mix, depending on your group's **norms** and the **incentives** your instructor set.

The four species and value (relative prices)

Relative value (highest to lowest in this model):

Species	Value in the model
Large goldfish	Highest value per fish (gold and large)
Small goldfish	High, but less than large gold (still gold)
Large silverfish	Less than either gold (silver), but large > small for silver
Small silverfish	Lowest value per fish

Gold types are worth more than **silver** types. **Large** types are worth more than **small** types, within the same “metal” group.

Prices and seasons: The **price per fish** (what your group “earns” for each type caught) **risks in season 2 and again in season 3** relative to the previous season, so the **temptation** to harvest heavily can **increase** over time unless your group's strategy accounts for the **reproduction** step.

Reproduction rule (after each season, before the next)

Work **separately for each of the four species:**

1. Count how many fish of that species were **not** caught (the fish **left in the pool** after that season's harvest). Call this **N** (a whole number).

2. For that species, the pool **gains** new fish. **Divide N by 2 and round down** to a whole number (in math this is the *floor* of $N/2$). That number is how many **new** fish of that species you **add**.
3. Example: $N = 5 \rightarrow 5 / 2 = 2.5 \rightarrow 2$ new fish (round down). New total = $5 + 2 = 7$ before the next season.
4. The **caught** fish are **removed** from the pool. They do **not** count toward reproduction in this model.

If your instructor uses a **different** reproduction rule, **follow their version**.

Connection to “evolution” in BIOL-8

Evolution is not only long-term **genetic** change. Biologists also study how **ecological and social** conditions **select** for behaviors, norms, and structures (e.g. cooperation, punishment of cheaters) that affect who survives and **reproduces**—including in **H. sapiens**. This lab is a **classroom analogy**: no real evolution in the fish, but real discussion about **incentives, group outcomes**, and **sustainability**—concepts that sit next to **natural selection** in a broader life-science view.

Materials (check with your instructor)

- **Shared pool** of countable units for four “species” (e.g. colored chips, small cards, or tallies) — **large gold, small gold, large silver, small silver**
 - A way to record **start, catch, left, reproduction add**, and **end** counts **per species, per season**
 - **Price list** for each **species** in **each season** (instructor or board)
 - This handout and writing tools
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Procedure

Before season 1

1. **Form groups** and write **all members’ names** in the roster below. You will use **three fishing seasons**.
2. For **each season**, **one** group member is the **fisher** for that season (someone who was not the fisher the previous time). **Rotate** so **three different people** fish in seasons 1–3 (if the group has fewer than three people, your instructor will say how to handle rotation).
3. Copy the **starting population** for each **species** from the instructor (or the board) into **Table 1** under **Start (Season 1)**.

4. Copy the **price per fish** for **each species in season 1** into **Table 2** (or record “see board” and keep your own clear notes).

Each season (repeat for seasons 1, 2, and 3)

1. The **fisher** for that season “catches” by **taking** units from the pool as allowed by the rules your instructor gives (e.g. **maximum** number of fish, or a **time** limit, or a **fixed** haul). Record **only** the **number caught per species** in **Table 1** in the **Catch** row for that season.
2. **After** the catch, compute **left** in the pool: **Start – catch** (per species) for that season. Enter under **Left after catch**.
3. **Reproduction:** For **each** species, compute **add** = (**Left after catch**) / 2, **rounded down** to a whole number. New count before the next season = **Left after catch + add**. Enter under **Repro add** and **End (after repro)**. The **End** row is the **Start** for the next season.
4. **Earnings for this season:** For each species, (**catch**) × (**price for that species in this season**). Sum to get **group earnings** for the season. Use **Table 2**; your instructor will give **season 2 and season 3** prices (higher than before).
5. The **next** group member **fishes** the following season, starting from the new **Start** row.

After season 3

- Total **cumulative** earnings if your instructor requests it.
 - Complete **Table 1** and **Table 2**, then the **Analysis** and **Conclusion** sections.
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Data

Abbreviations: LG = large gold, SG = small gold, LS = large silver, SS = small silver.

Group roster (fill in before Season 1)

Group roster

#	Role	Name
1		
2		
3		
4		

Table 1 — Population and catch (all three seasons)

Starting populations (from instructor) go in the **Start — Season 1** row below.

Table 1 — Season 1 population and catch

#	Large gold (LG)	Small gold (SG)	Large silver (LS)	Small silver (SS)
1				
2				
3				
4				
5				

Season 2 — *Fisher (name):*

Table 1 (continued) — Season 2

#	LG	SG	LS	SS
1				
2				
3				
4				
5				

Season 3 — *Fisher (name):*

Table 1 (continued) — Season 3

#	LG	SG	LS	SS
1				
2				
3				
4				
5				

Table 2 — Prices by season, then earnings

Copy **price per fish** from the instructor. **Earnings** for a season = sum over species of (catch that season) $\times$ (price that season for that species).

Table 2 — Price per fish by season

#	Species	Price — Season 1	Price — Season 2	Price — Season 3
1				
2				
3				
4				

Table 2 — Earnings by season

#	Season	Earnings (show work or subtotal)
1		
2		
3		
4		

Analysis

Answer in **complete sentences**. Use your **tables** for any question that refers to your group's numbers.

1. Tragedy of the commons. In two or three sentences, did your group's pool **trend toward depletion, stability, or recovery** for one or more species? Name **one** choice that made that outcome more likely.

2. Evolution of incentives. Prices rose each season. How did that affect what you think the "best" **short-term** move was, versus a **long-term** group goal?

3. Game theory. Name **one** behavior that acted like **cooperation** and **one** that acted like **defection** (or self-interest only), whether or not you used those words during the game.

4. Reproduction rule. For **one** species, show with numbers: **left after catch, repro add** (left / 2, **round down**), and **end** after you add. Did the rule ever **increase** the stock enough that your group was glad some fish were **left in the pool**?

Conclusion

5. Human biology in one sentence: Why might a **lecture** on evolution in humans include a **resource** and **incentive** story like this, not only DNA and fossils?

Lab 18 — Evolution. Tragedy of the commons and game theory via a three-season fishing simulation.